

Updated E-Learning Portal Database Schema Review

Prepared after reviewing the project description and the current database schema. This document identifies the recommended schema, relationships, and the changes that should be made before implementing Entity Framework Core Code First.

Recommended Tables (24 Core Tables)

1. Branch
2. Role
3. User
4. Category
5. DifficultyLevel
6. MasterCourse
7. SubCourse
8. Topic
9. Video
10. LearningMaterial
11. Assignment
12. StudentAssignment
13. MCQQuestion
14. MCQAttempt
15. StudentTopicProgress
16. CourseEnrollment
17. SubscriptionPlan
18. SubscriptionCourse
19. StudentSubscription
20. Payment
21. Invoice
22. Certificate
23. Rating
24. Feedback

Relationships

- Branch (1) -> (*) Users
- Role (1) -> (*) Users
- Branch (1) -> (*) MasterCourses
- Category (1) -> (*) MasterCourses
- DifficultyLevel (1) -> (*) MasterCourses

- MasterCourse (1) -> (*) SubCourses
- SubCourse (1) -> (*) Topics
- Topic (1) -> (*) Videos
- Topic (1) -> (*) LearningMaterials
- Topic (1) -> (*) Assignments
- Topic (1) -> (*) MCQQuestions
- Assignment (1) -> (*) StudentAssignments
- User(Student) (1) -> (*) MCQAttempts
- User(Student) (1) -> (*) StudentTopicProgress
- User(Student) (1) -> (*) CourseEnrollments
- SubscriptionPlan (1) -> (*) SubscriptionCourses
- SubscriptionPlan (1) -> (*) StudentSubscriptions
- Payment (1) -> (1) Invoice
- MasterCourse (1) -> (*) Ratings
- MasterCourse (1) -> (*) Feedback
- MasterCourse (1) -> (*) Certificates

Changes Required in Existing PDF

Users: Split FullName into FirstName & LastName. Add ProfileImage, Email verification fields, ResetPassword fields, ModifiedDate.

Branch: Add State and Logo.

MasterCourses: Remove stored Price. Calculate from SubCourse prices. Add CreatedDate.

SubCourses: Add Thumbnail and CreatedDate.

Topics: Add Description.

Videos: Use VideoPath instead of URL. Add Thumbnail and VideoSize.

LearningMaterials: Rename FileURL to FilePath. Add Title and CreatedDate.

Assignments: Add Description and CreatedDate.

StudentAssignments: Add Marks, Remarks and ReviewedDate.

MCQQuestions: Add Marks.

MCQAttempts: Add TotalMarks.

StudentTopicProgress: Add AssignmentReviewed flag.

CourseEnrollments: Add Status.

SubscriptionPlans: Add CreatedDate.

StudentSubscriptions: Add Status.

Payments: Add PaymentFor, CourseId, SubscriptionId, TransactionId.

Invoices: Add GST if applicable.

Certificates: Add CertificateUrl.

Ratings: Restrict values to integers 1-5.

Feedback: Rename Feedback to Comment for clarity (optional).

Recommendations

Keep the schema at 24 core tables. Do not add Notification, EmailLog, or PurchaseHistory tables for the initial one-week project unless the team explicitly wants production-level features. Configure all foreign-key relationships using Fluent API in ApplicationDbContext with DeleteBehavior.Restrict where cascading deletes could cause data loss.